

What Really Drives Our Power System Models? Identifying the Assumptions That Shape ERCOT Power Futures

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Abstract

Long-term power-system decarbonization studies increasingly rely on large-scale capacity-expansion models, yet many analyses remain deterministic despite substantial uncertainty in technology costs, fuel prices, and electricity demand. This study applies a structured uncertainty workflow in the Regional Energy Deployment System (ReEDS) model to identify which uncertain assumptions most strongly influence renewable curtailment in ERCOT (Texas), an operability-focused indicator of variable renewable energy integration challenges. We define data-grounded uncertainty bounds from published trajectories and use a Morris (elementary effects) screening design to rank influential drivers based on the mean absolute elementary effect (μ^*). The analysis is conducted through 2050 using ReEDS' two-year time step. The resulting rankings support more transparent interpretation of ReEDS-based planning results by focusing attention on the uncertainties that most strongly drive curtailment outcomes. Preliminary results from the first completed Morris trajectory (40 of 200 runs) indicate that ERCOT curtailment is most sensitive to a small set of economic and performance drivers—especially natural-gas price conditions, renewable output assumptions (e.g., wind capacity-factor scaling), and Gas-CC operating cost/efficiency parameters—implying that scenario development and interpretation can be prioritized around these levers; future work will complete the full 200-run design to confirm ranking stability and assess potential interactions.

Keywords

ReEDS; Morris method; elementary effects; global sensitivity analysis; uncertainty characterization; renewable curtailment

1. Introduction

Long-term energy and climate targets require a rapid shift away from fossil-based energy supply and toward low-carbon technologies across the economy [1]. This transition involves coordinated investment decisions in generation, conversion, storage, and network infrastructure, with assets that persist for decades and interact across sectors [1]. Consequently, evaluating alternative decarbonization pathways—in terms of cost, feasibility, and emissions outcomes—has become a core motivation for energy-system planning and modelling studies [1–3].

Deep decarbonization pathways typically feature high shares of variable renewable energy (VRE) and widespread electrification, increasing the importance of representing temporal variability (e.g., hourly correlations between VRE output and load) and geographical heterogeneity (e.g., resource quality, siting limits, and network constraints) [2, 4]. These requirements have motivated the use of large-scale capacity-expansion and energy system optimisation models with explicit representation of generation portfolios, transmission expansion, and operational constraints [2]. In the United States, the Regional Energy Deployment System (ReEDS) model is widely used for this purpose: ReEDS is a long-term, least-cost capacity-expansion model of the U.S. power sector that endogenously selects generation, storage, and transmission investments while accounting for regional resource quality and system adequacy constraints [5]. Increasing spatial and temporal fidelity in such models can improve realism for VRE-dominated systems, but also increases data requirements and computational burden [4, 6].

Although these modelling advances support policy-relevant planning, many energy-system studies remain predominantly deterministic, even though key inputs such as technology costs, fuel prices, and demand trajectories are uncertain because they rely on estimation and forecasting [2, 7]. This uncertainty is particularly consequential in capacity-expansion settings like ReEDS, where results inform long-lived, capital-intensive investment decisions that can create lock-in effects [1, 2, 8], and where policy-facing conclusions depend on transparent documentation of the assumptions driving model outcomes [7]. These issues are especially relevant for regional planning contexts such as

ERCOT (Texas), where high VRE shares and rapid load growth can amplify integration challenges and increase the value of identifying the key uncertain drivers that shape model outcomes.

Historically, uncertainty in energy modelling has often been explored through deterministic baselines augmented by a limited number of alternative scenarios or one-at-a-time sensitivity tests [2]. While these approaches can provide intuition, they scale poorly with the dimensionality of uncertainty and can miss interaction effects among uncertain drivers [7]. A more general approach is global sensitivity analysis (GSA), which evaluates uncertain inputs simultaneously and attributes variation in outputs to the uncertain inputs [7]. In practice, variance-based GSA methods can be computationally expensive for large-scale energy models with many uncertain inputs [7], motivating screening strategies that can efficiently identify which parameters matter most for outcomes of interest [7, 9, 10].

A widely used screening approach is the Elementary Effects method (commonly referred to as the Morris method), which provides a computationally efficient way to rank influential inputs and support workflows such as factor prioritization (identifying influential drivers) and, where appropriate, factor fixing (identifying uninfluential drivers that can be held constant) [7]. Methodological guidance for ESOMs emphasizes that the credibility of any sensitivity ranking depends strongly on how uncertain inputs are defined; specifically, uncertainty characterization should precede the sensitivity analysis because poorly justified or generic uncertainty ranges can produce misleading importance rankings [7, 9]. This logic is particularly relevant for power-system planning studies, where uncertainties in techno-economic assumptions (e.g., costs and performance), exogenous drivers (e.g., demand), and market variables (e.g., fuel prices) jointly influence investment and dispatch decisions over long horizons [2, 9]. Taken together, the literature motivates structured workflows that (i) define defensible uncertainty ranges grounded in data sources and modelling conventions, and (ii) apply scalable screening methods to identify the most influential uncertain drivers in the specific model and application context [7, 9].

Against this backdrop, this study positions uncertainty characterization and Morris-style screening as a practical pathway for improving the transparency and robustness of ReEDS-based decarbonization analysis for ERCOT (Texas) [7]. By identifying which uncertain assumptions most strongly shape ReEDS outputs, the goal is to focus scenario design and data-improvement efforts on the drivers that matter most, while clearly communicating how conclusions depend on uncertain inputs [7, 9]. Concretely, we vary key exogenous drivers in ReEDS using a Morris (elementary effects) design: technology cost and performance parameters (capital cost, fixed O&M, variable O&M, and efficiency/heat rate), electricity demand, and natural gas prices. We then quantify how these uncertainties propagate to a single operability-focused outcome—renewable curtailment in ERCOT—which serves as an interpretable indicator of integration and flexibility challenges in VRE-rich pathways [4]. Preliminary results from the first completed Morris trajectory (40 of 200 runs) already point to a clear pattern: ERCOT curtailment responds most strongly to a small subset of uncertainties—natural-gas price conditions, renewable output assumptions (especially wind capacity-factor scaling), and Gas-CC cost/efficiency parameters. This early signal motivates prioritizing these levers in subsequent scenario design and interpretation, while the full 200-run design will be completed to confirm the stability of the rankings and to characterize potential interaction effects.

2. Methodology

2.1. Overview

We conduct a screening-based global sensitivity analysis (GSA) to identify which uncertain exogenous drivers most strongly influence renewable curtailment in Texas (ERCOT) using the ReEDS power system optimization framework. The methodology consists of four linked stages: (i) selecting candidate uncertain inputs for screening, (ii) deriving data-grounded uncertainty ranges from published sources, (iii) generating a Morris (elementary effects) sampling design and executing the corresponding ReEDS runs, and (iv) computing elementary-effects sensitivity indices to rank factor importance [11]. The ReEDS model is a long-term, least-cost capacity-expansion model of the U.S. power sector that endogenously selects generation, storage, and transmission investments while representing regional resource quality and adequacy-relevant constraints [5, 12]. Uncertainty ranges for technology parameters are derived from the NREL Annual Technology Baseline (ATB) [13], while electricity demand and natural gas price ranges are derived from the U.S. EIA Annual Energy Outlook (AEO) cases [14]. The overall workflow is summarized in Figure 1.

2.2. Pre-screening: candidate uncertain inputs

To evaluate how key exogenous drivers shape curtailment outcomes in ERCOT, we parameterize uncertainty in three categories: (i) technology cost and performance metrics, (ii) electricity demand, and (iii) natural gas price. This struc-

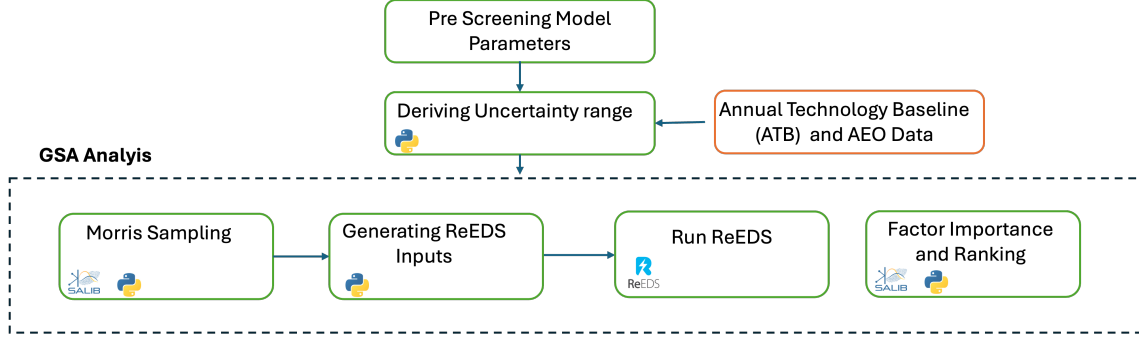


Figure 1: End-to-end workflow for Morris screening in ReEDS for the ERCOT study scope. Candidate inputs are selected, uncertainty ranges are derived from ATB and AEO data, a Morris design is generated, sampled values are written into ReEDS input files, ReEDS scenarios are executed, and elementary-effects indices are computed to rank influential drivers of renewable curtailment.

ture is intended to capture dominant sources of variability that affect capacity expansion and dispatch decisions: technology competitiveness (cost and performance), the magnitude of the load that must be served, and fuel-market conditions that influence operating costs and the relative economics of thermal generation [11]. The screened technology parameters include overnight capital cost (capcost), fixed O&M (fom), variable O&M (vom), and heat rate/efficiency (for thermal technologies), along with renewable performance parameters where applicable (e.g., capacity-factor multipliers or improvement factors) [5, 13].

2.3. Deriving data-grounded multiplier ranges from ATB and AEO

We define uncertainty ranges using *multipliers* applied to a baseline (reference) time series. The idea is to keep the baseline trajectory shape over time, but allow its level to move up or down within bounds supported by published “high” and “low” cases. For any input series (e.g., a technology capital cost trajectory, a demand trajectory, or a fuel price trajectory), let $x_{\text{ref}}(t)$ denote the reference series and $x_{\text{tar}}(t)$ denote a bounding series (high or low). We estimate a single scalar multiplier m that best scales the reference series to match the bounding series over the policy-relevant years (here, $t \geq 2025$) using a least-squares fit:

$$m^* = \arg \min_m \sum_{t \in \mathcal{T}} (m x_{\text{ref}}(t) - x_{\text{tar}}(t))^2, \quad \mathcal{T} = \{t : t \geq 2025\}. \quad (1)$$

The fitted multipliers then define lower and upper bounds as

$$x_{\text{low}}(t) = m_{\text{low}} x_{\text{ref}}(t), \quad x_{\text{high}}(t) = m_{\text{high}} x_{\text{ref}}(t), \quad (2)$$

which yields transparent, data-grounded uncertainty ranges that are well suited to Morris screening [11].

For technology cost and performance parameters, we use NREL ATB trajectories: the ATB *moderate* case is the reference, and the ATB *conservative* and *advanced* cases provide the bounding (low/high) trajectories [13]. For electricity demand and natural gas price, we apply the same approach using the U.S. EIA AEO *reference*, *low*, and *high* cases [14]. Figure 2 illustrates the procedure for offshore wind (ATB 2024, floating) by showing the moderate, advanced, and conservative trajectories and the fitted multiplier envelope over years ≥ 2025 .

2.4. Morris (elementary effects) sampling design

We apply the Method of Morris (Elementary Effects) to screen influential inputs with a computational budget that scales linearly with the number of factors [11, 15–17]. Each factor is mapped to the normalized hypercube $[0, 1]$ and discretized on a p -level grid; we use $p = 6$ and the standard Morris step size $\Delta = \frac{p}{2(p-1)}$ [15, 16]. A Morris trajectory contains $(k + 1)$ design points generated by perturbing one factor at a time by $\pm\Delta$, and repeating this for r trajectories yields $N = r(k + 1)$ model runs; here we use $r = 5$ and a fixed random seed for reproducibility [15, 17]. Normalized samples are linearly scaled to the physical multiplier bounds [17, 18].

2.5. Generating ReEDS inputs (automated run-file creation) and execution

After sampling the uncertainty space with the Morris design, the practical task is to turn each sampled point into a ReEDS run that can be executed without manual edits. We therefore implement an automated *input-generation*

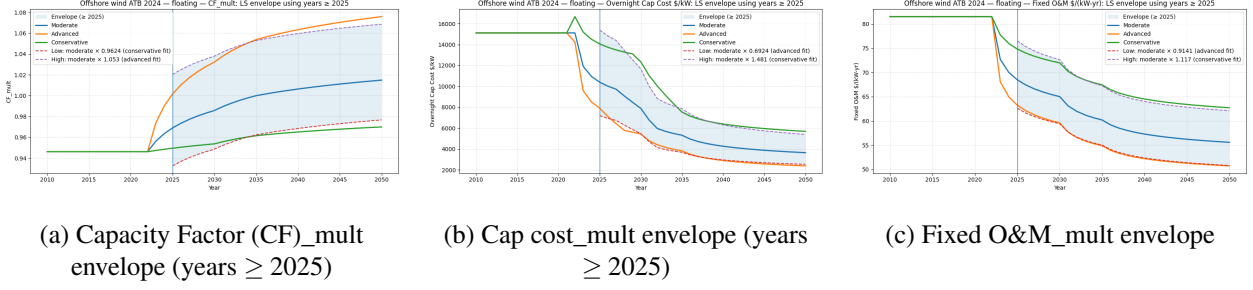


Figure 2: Example of multiplier-range derivation for offshore wind (ATB 2024, floating). The shaded region shows the fitted uncertainty envelope for years ≥ 2025 , constructed by scaling the reference (moderate) trajectory using least-squares multipliers fitted to the bounding trajectories (advanced and conservative). The resulting multipliers provide the lower and upper bounds used in the Morris screening design.

step that (i) applies the sampled multipliers to the appropriate baseline input datasets and (ii) writes the results in the exact CSV formats and file locations expected by the ReEDS preprocessing pipeline [5, 12]. This ensures that every run is “ReEDS-compatible” by construction and can be launched in batch. All scenarios are executed for the ERCOT (Texas) balancing-area study scope and solved over the long-term planning horizon through 2050, with results reported on ReEDS’ two-year time step [5, 12].

2.6. Ranking and factor importance (Morris indices)

For each run, we extract the output metric of interest (renewable curtailment in ERCOT) and compute Morris *elementary effects* (EE) for every factor as the finite-difference change in output induced by a one-factor perturbation along each trajectory [15–17]. Let $f(\mathbf{x})$ denote curtailment for an input vector $\mathbf{x}$ on the normalized grid and let Δ be the Morris step size. The elementary effect of factor i at point $\mathbf{x}$ is

$$EE_i(\mathbf{x}) = \frac{f(\mathbf{x} + \Delta \mathbf{e}_i) - f(\mathbf{x})}{\Delta}, \quad (3)$$

where $\mathbf{e}_i$ is the unit vector for factor i [15]. Repeating this across r trajectories yields a sample of elementary effects for each factor, which are summarized using the mean absolute elementary effect

$$\mu_i^* = \frac{1}{r} \sum_{j=1}^r |EE_i^{(j)}|, \quad (4)$$

as the primary measure of overall influence, since it avoids cancellation when effects change sign [11, 16, 17]. We additionally report the standard deviation

$$\sigma_i = \sqrt{\frac{1}{r-1} \sum_{j=1}^r (EE_i^{(j)} - \bar{EE}_i)^2}, \quad (5)$$

as an indicator of potential nonlinearity and/or interactions [11, 16]. Factors are ranked by descending μ^* to identify the dominant uncertain drivers of ERCOT curtailment; where σ is large relative to μ^* , results are interpreted as reflecting stronger nonlinearities or interaction effects and motivate follow-on analysis with more detailed uncertainty quantification if needed [11, 17].

3. Results: Preliminary Morris screening of ERCOT curtailment

We applied the Method of Morris (elementary effects) to screen which uncertain inputs most influence *total curtailment in ERCOT*, where the outcome is defined as the sum of curtailment across all modeled years from the ReEDS curtailment output (ReEDS solved for ERCOT). The uncertainty set includes $k = 39$ factors spanning technology cost/performance multipliers, a Texas electricity-demand multiplier, and a natural-gas price multiplier for West South Central. Using $p = 6$ grid levels and $r = 5$ trajectories yields a design size of $r(k + 1) = 200$ model evaluations. At the time of writing, only 40 runs have completed, corresponding to one full trajectory (trajectory 0). Therefore, the results below should be interpreted as an initial screening and are expected to stabilize as additional trajectories complete (e.g., σ is not yet estimable with a single trajectory; $n_{EE} = 1$ per factor).

Figure 3 shows the preliminary ranking using normalized μ (mean absolute elementary effect in a normalized $[0, 1]$ input space). The dominant driver of ERCOT curtailment in the completed trajectory is the natural-gas price multiplier,

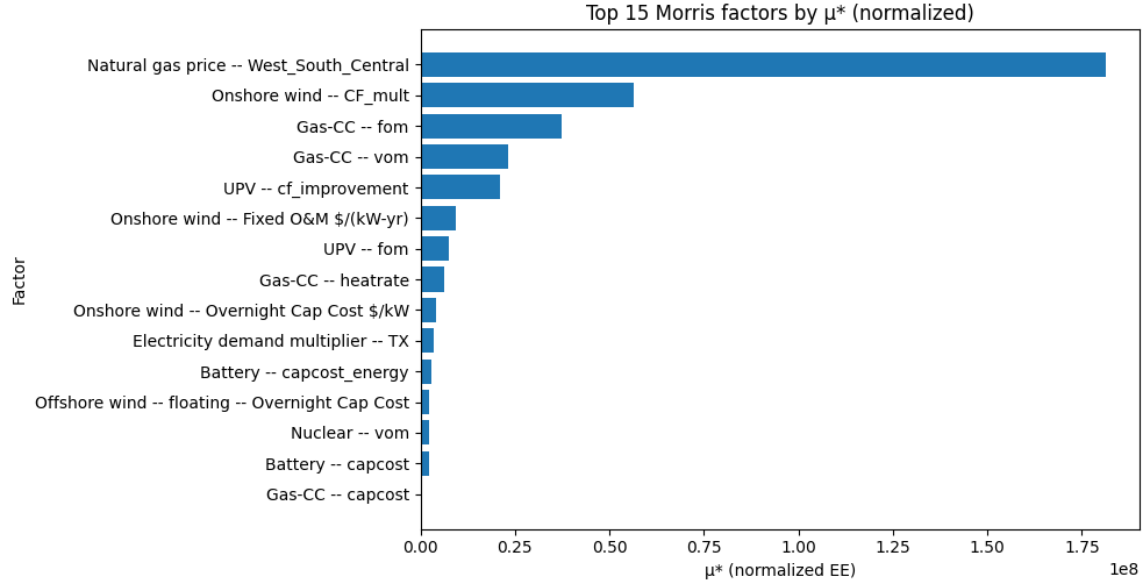


Figure 3: Preliminary Morris screening results for ERCOT total curtailment. Bars show the top 15 factors ranked by normalized μ from the single completed Morris trajectory (trajectory 0). Results are preliminary pending completion of the full $r = 5$ trajectory design (200 runs).

followed by the onshore wind capacity-factor multiplier and several Gas-CC operating-cost/performance parameters (FOM, VOM, heat rate). Additional influential factors include PV capacity-factor improvement, onshore wind fixed O&M and overnight capital cost, and the Texas electricity-demand multiplier. It can be seen that curtailment is governed by relative dispatch/build economics and system flexibility, so fuel-price conditions and renewable output assumptions strongly shift both the supply mix and the frequency of surplus VRE generation. Several coal-related parameters exhibit near-zero effects in this trajectory, which may reflect local insensitivity (non-marginal technologies under these conditions) and/or interactions that require multiple trajectories to reveal.

4. Discussion and Next Steps

The preliminary Morris screening (one completed trajectory; 40 of 200 runs) shows the benefit of running a screening method *before* large scenario ensembles: it expands the explored input space in a structured way and improves interpretability by directly linking changes in total ERCOT curtailment to specific drivers. Even at this stage, the ranking suggests that fuel-price conditions and variable renewable output assumptions are key levers affecting curtailment, providing a clear basis for prioritizing subsequent analyses.

Next steps are to (i) complete the remaining trajectories to stabilize μ rankings and enable estimation of interaction/nonlinearity indicators (e.g., σ), (ii) develop a full DACE workflow for ReEDS using space-filling designs focused on the most influential factors, and (iii) build surrogates (emulators) for ReEDS outputs to enable rapid exploration and to support linking with other models through a computationally efficient interface.

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